



DEPARTMENT OF COMPUTER SCIENCE ENGINEERING

Rajiv Gandhi University of Knowledge Technologies – Nuzvid

Nuzvid, Eluru, Andhra Pradesh – 521202.

HOSTEL MONITORING SYSTEM

A Project Progress Report

Submitted in partial fulfillment for the degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING

Submitted by

M. JOSHI (N200395)

S. GANESH (N200197)

N. SYAM PRASAD (N200773)

M. UMA MAHESWARI (N200289)

A. TANOOSHA (N201048)

Under the Esteem Guidance of

Mrs. K. Krishna Priya



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CERTIFICATE OF COMPLETION

This is to certify that the work entitled, “**Hostel Monitoring System**” is the bonafied work of **M. JOSHI (ID No: N200395), S. GANESH (ID No: N200197), N. SYAM PRASAD (ID No:N200773), M. UMA MAHESWARI (ID No:N200289), A. TANOOSHA (ID No:N201048)** carried out under my guidance and supervision for 3rd year project of **Bachelor of Technology** in the department of Computer Science and Engineering under RGUKT IIIT Nuzvid. This work is done during the academic session December 2024 – April 2025, under our guidance.

Mrs. K. Krishna Priya

Assistant professor,

Department of CSE,

RGUKT Nuzvid.

Mrs. S Bhavani

Assistant Professor,

Head of the Department,

Department of CSE,

RGUKT Nuzvid.



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CERTIFICATE OF EXAMINATION

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Mrs. K. Krishna Priya

Assistant professor,

Department of CSE,

RGUKT Nuzvid.

Mr. Jaya Krishna

Assistant Professor,

Department of CSE,

RGUKT Nuzvid,



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Rajiv Gandhi University of Knowledge Technologies – Nuzvid

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DECLARATION

M. JOSHI (ID No: N200395), S.GANESH (ID No: N200197), N. SYAM PRASAD (ID No:N200773), M. UMA MAHESWARI (ID No:N200289), A. TANOOSHA (ID No:N201048) hereby declare that the project report entitled “**Hostel Monitoring System**” done by us under the guidance of **Mrs. Krishna Priya**, Assistant Professor is submitted for the partial fulfillment for the award of degree of Bachelor of Technology in Computer Science and Engineering during the academic session December 2024 – April 2025 at RGUKT - Nuzvid.

We also declare that this project is a result of our own effort and has not been copied or imitated from any source. Citations from any websites are mentioned in the references. The results embodied in this project report have not been submitted to any other university or institute for the award of any degree or diploma.

Date: 18-04-2025

Place: Nuzvid

N200395 M. JOSHI

N200197 S. GANESH

N200773 N. SYAM PRASAD

N200289 M. UMA MAHESWARI

N201048 A. TANOOSHA

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We are extremely grateful for the confidence bestowed in us and entrusting our project entitled **“Hostel Monitoring System”**.

We express gratitude to **Mrs. S. Bhavani (HOD of CSE)** and other faculty members for being source of inspiration and constant encouragement which helped us in completing the project successfully.

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1. Abstract

The **Hostel Monitoring System** is a smart solution made for university hostels to help manage complaints and bus bookings in an easy and fast way. We built a web dashboard for admins and caretakers, and a mobile app for students.

Students can log in using their college email, raise complaints through the app, and track their status. These complaints are instantly sent to the admin and the concerned hostel caretaker, who can respond by adding comments and solving the issue quickly. The bus booking feature was added to replace the old, manual process that took a lot of time and effort. Now, the admin can send a form asking students if they want to travel by the university bus and which city they prefer. Based on the responses, the admin can create proper bus routes with seat details and ticket prices. Students can then book tickets through the app and make payments using PhonePe. To avoid double bookings, we used Redis to handle seat locks during high-traffic times.

This project helps hostel staff work faster, improves communication, and makes things more convenient for students. It can also be used in other hostels and institutions that face similar problems.

2. Introduction

A Smarter Way to Manage Hostels

Managing hostel complaints and holiday travel manually used to take a lot of time and effort. Students often didn't get proper updates on their complaints, and bus bookings during holidays were handled manually, which caused delays, and extra work for staff. Our project solves these problems by bringing everything into one smart digital system that is fast, simple, and transparent for both students and hostel management.

2.1 Project Overview

We built a Hostel Monitoring System that has two parts — a web dashboard for admins and caretakers, and a mobile app for students.

Students can:

- Raise complaints
- Book holiday buses

Admins and caretakers can:

- Track and respond to complaints
- Collect travel plans and manage buses based on student needs

2.2 Why This Project?

Hostel management was done manually before — it caused delays, confusion, and a lot of paperwork. Students and Staff had no easy way to get help or manage arranging buses.

Our platform:

- Makes it easy for students to report problems
- Helps staff respond faster and better
- Makes holiday bus bookings quick, smooth, and fair
- Reduces confusion and human error

2.3 What Does the Platform Do?

- **Complaint Management**
 - Students raise complaints through the app
 - Admin and caretaker get notified instantly
 - They can reply, add comments, and solve the issue faster
- **Bus Booking System**
 - Admin shares a form asking students where they want to travel
 - Based on the responses, bus routes are created
 - Students book tickets, pay with PhonePe, and secure their seats
 - We use Redis to avoid double bookings during high demand

2.4 What's Special About This Project?

- **Fast & Transparent** : No more waiting for updates — everything is tracked and clear.
- **Easy Communication** : Admins and caretakers can reply directly on the complaint thread.
- **Smart Bus Booking** : Decisions are based on student feedback. It's fair and efficient.
- **Safe Payments** : Integrated with PhonePe for smooth and secure ticket booking.
- **Real-Time Seat Locking** : Stops two people from booking the same seat at the same time.

2.5 Real-World Application

This project is built specifically for use in our university hostels and is ready to be used with real-world data and users. It simplifies how students and hostel staff interact, making everyday processes faster, clearer, and more efficient.

- **University Hostels**

The system is designed to work perfectly in our university's hostel environment. It manages complaints, improves communication between students and caretakers, and tracks complaint status in real time.

- **Holiday Travel Booking for Students**

With just a few clicks, students can view and book bus tickets during holidays. Admins can easily collect student responses, view insights, and decide which cities to arrange transport to, saving hours of manual work.

- **Admin & Caretaker Dashboard**

Staff can quickly respond to student issues, track progress, and keep things organized. No need for paperwork or phone calls — everything is handled through one dashboard.

- **Real-Time Monitoring and Updates**

All actions, from booking seats to resolving complaints, are updated instantly and stored securely. It works with live student data and is ready to be implemented anytime.

This system is not just a prototype — it's a practical solution made for the real world and is ready to be deployed in our university.

3. Related Works/ Existing Systems

Our project is unique and fully developed from scratch to solve real issues in our university hostel system. While there are no existing platforms that do exactly what our system does, we looked at the current manual methods used in our campus for handling complaints and bus bookings. These methods helped us understand what needs to be improved.

Existing Complaint System :

Currently, there is no digital way for students to raise and track complaints. The process is:

- A physical complaint box is placed in hostels where students write complaints on paper and drop them in.
- This box is opened only once a month, which means complaints are often delayed or ignored.
- For urgent issues, a complaint book is maintained by the caretaker. Students must go in person, find the book, and write their complaint manually.

This system is time-consuming, lacks updates or tracking, and is hard to manage efficiently.

Existing Bus Booking Process :

During holidays, the university arranges buses to different cities for students. Currently, the process is:

- Staff members send out Google Forms to collect responses about where students want to travel.
- Based on the responses, they manually arrange buses.
- Students are asked to pay in person, and their names are written down in notebooks.
- Receipts are then handed over manually to each student.

This leads to confusion, last-minute changes, and a lot of effort for both students and staff.

How Our Project is Different:

Our system replaces all of this with a smart, digital platform that handles everything from start to finish:

- Students can raise complaints anytime online, and get real-time updates.

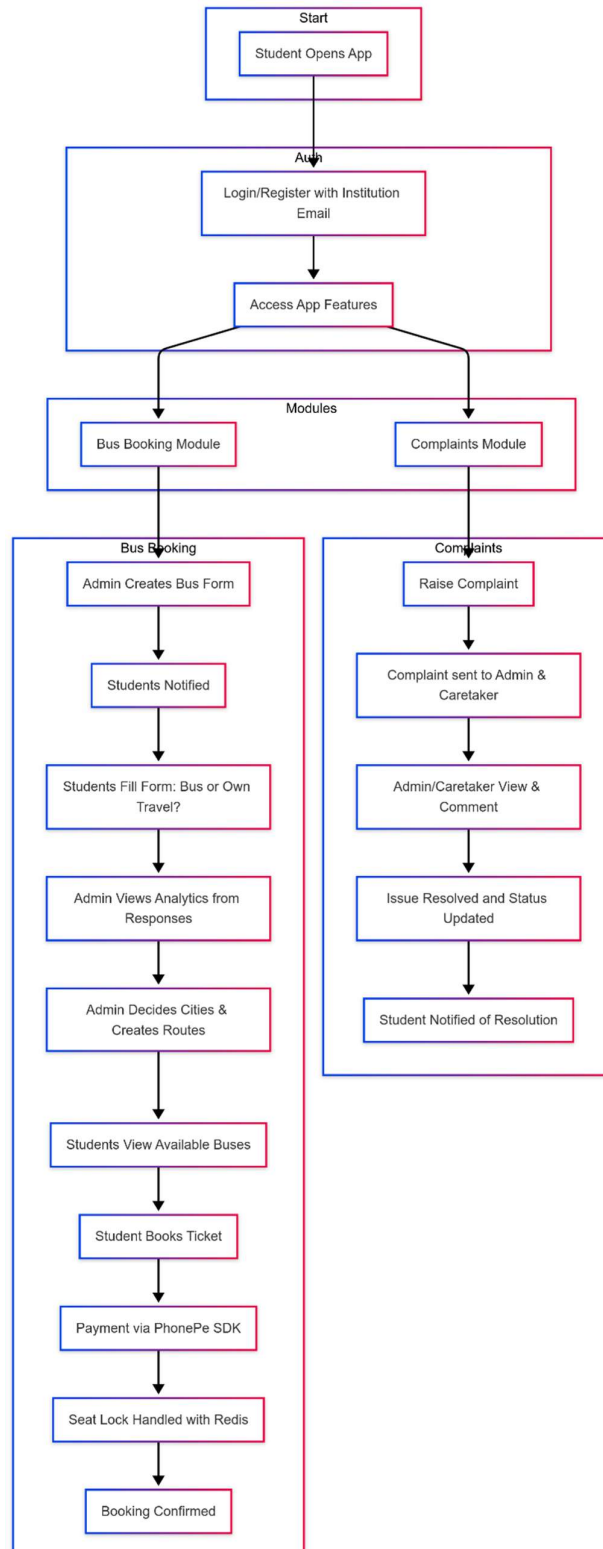
- Caretakers and admins can reply and track issues with no paperwork.
- Bus bookings can be done within seconds through the dashboard with digital seat allocation and confirmation.
- No need for physical receipts or manual cash collection — everything is transparent and fast.

By using modern technology, our platform saves time, avoids confusion, and makes the hostel experience smoother for everyone.

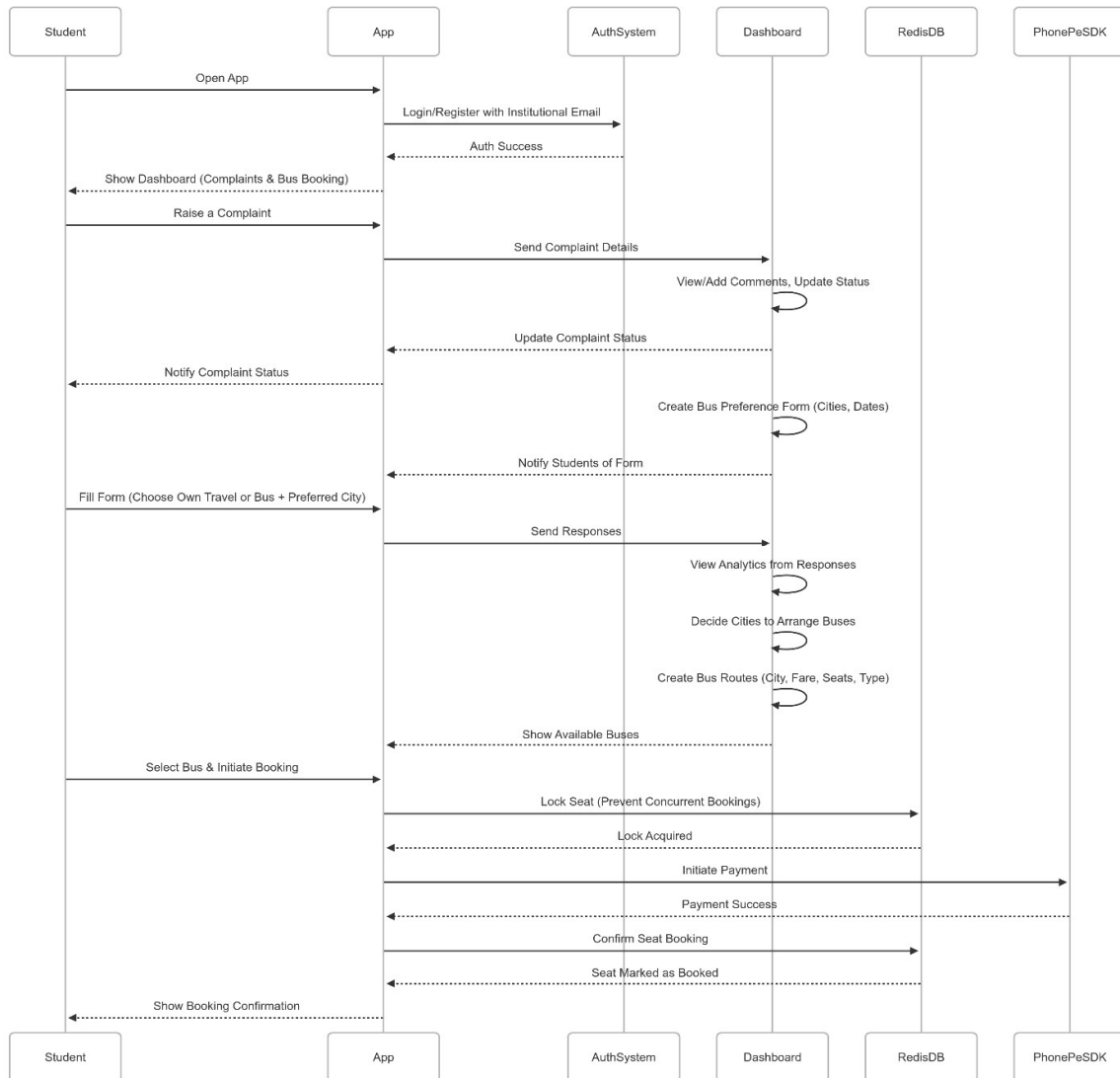
4. Proposed Method

4.1 System Flow Diagram:

Flow Chart:



Sequence Diagram:



4.2 Module-Wise Breakdown & Data Flow:

User Authentication & Role Management:

Purpose:

Secure access control for different users like students and admins.

Functionality:

- Students log in through the **mobile app**.
- Admins access the system via the **admin portal**.
- Authentication is implemented using **JWT tokens** for secure session handling.
- Roles and permissions:
 - **Students:** Can lodge complaints, view status, book buses.
 - **Admins:** Can manage complaints, bookings, bus routes, and payments.

Complaint Management:**Purpose:**

Digitize the process of handling hostel-related complaints.

Functionality:

- Students raise complaints with category, urgency, and description.
- Complaints are stored in database and categorized as:
 - Pending → In Progress → Resolved
- Admins can update status and respond to complaints via their dashboard.
- Historical logs and timestamps are maintained for tracking.

Smart Bus Booking:**Purpose:**

Enable seamless travel planning for holidays.

Functionality:

- Students view available buses, timings, and book seats through the app.
- Admins create and manage bus routes, schedules, and seat capacities via the portal.
- Booking status (confirmed/cancelled) is updated in real-time.
- Students receive digital confirmation instead of manual receipts.

Real-Time Seat Locking & Concurrency (Redis):

Purpose:

Prevent double-booking of bus seats by handling concurrent bookings.

Functionality:

- When a student selects a seat:
 - It is **locked in Redis cache** for a limited time (5 minutes).
 - If payment is not completed, the seat is auto-released.
- Ensures **fair and conflict-free** booking even with high traffic.

Payment Integration (PhonePe API):

Purpose:

Automate and secure the fee collection process for bus bookings.

Functionality:

- Students complete bookings by paying via **PhonePe** inside the app.
- Backend initiates and verifies payments using the **PhonePe Payment Gateway API**.
- Successful transactions trigger seat confirmation and generate digital receipts.
- All transactions are logged in **Database** for accountability.

5. Working of the Project

The project includes a mobile application for students, an admin dashboard for admins and caretakers, and a robust backend integrated with Redis for concurrency, MongoDB for persistence, Firebase for notifications, and PhonePe for secure payments.

5.1. Frontend Interfaces:

i. Student App (Flutter):

The mobile application is the primary interface for students, allowing them to:

- Register and log in using secure authentication
- Raise complaints to caretakers for resolution
- Book bus seats for travel during holidays
- Pay securely via PhonePe
- Receive real-time notifications about complaint status, bookings, and announcements

The app communicates with the backend using REST APIs and handles token-based authentication via JWT. State management ensures session persistence and smooth user navigation.

ii. Admin Dashboard (React.js):

Admins (admins and caretakers) access the system via a web dashboard. Key features include:

- Dashboard overview of complaints, bookings, and active buses
 - Manage complaint lifecycle (assign, resolve, close)
 - Configure bus details and timings
 - Send mass notifications or updates
- The admin portal is built using React.js and interacts with backend APIs for dynamic, real-time data access and updates.

5.2. Backend Services (Node.js + Express):

The backend is designed as a modular service-oriented architecture with Express routes grouped by functionality. It is responsible for handling business logic, interacting with data stores, and communicating with third-party services like Redis, PhonePe, and Firebase.

i. Authentication & User Session Management:

We went with JWT-based auth to keep things stateless and secure.

- Login/Register API: We validate the user, generate both access and refresh tokens, and send them to the client.
- Refresh Mechanism: A dedicated refresh-token endpoint issues a new access token when the old one expires.
- Middleware: Added an `auth.middleware.js` to protect routes and decode the token to fetch user roles (student/admin) from the payload.
- Role-based access: Certain routes like complaints or bus route creation are guarded so only admins can access them.

We didn't use sessions — just pure bearer tokens in headers, which works great for both mobile and web.

ii. Concurrency Booking using Redis:

One of the hardest parts was handling real-time seat locks. We chose Redis for its speed and TTL support.

How it works:

- When a student selects a seat, we check if a lock exists for: **"lock:busId:seatNo"**.
- If it doesn't, we create the lock in Redis with a 5-minute TTL.
- If the user doesn't pay in time, Redis auto-expires the lock — no cleanup needed.
- On successful payment, we:
 - Remove the Redis lock
 - Create a booking in MongoDB

To prevent race conditions, we used SETNX in Redis (atomic operation). This ensures two people can't lock the same seat simultaneously.

iii. Third Party APIs Integration:

- Notifications are triggered only when a bus form is created, prompting users to fill out travel details (how and where they will travel).

- Users subscribe to a specific topic related to the bus form for efficient mass notification.
- When the bus form is created, a notification is sent to the entire topic, ensuring all users are notified simultaneously.
- This approach eliminates the need to send individual notifications, reducing API calls and improving performance.

iv. PhonePe Payment Gateway API:

- The PhonePe API is integrated into the system to handle payments for bus bookings.
- Users are redirected to PhonePe for completing the payment process.
- Upon payment completion, the system receives status updates (success or failure) from PhonePe to update the booking status.
- The integration ensures secure payment handling and status tracking for a seamless user experience.

6. Experimental Setup

6.1 Technologies Used:

i. Frontend Development:

- **Flutter** – Used for building the cross-platform mobile app, ensuring a smooth and consistent user interface across both Android and iOS.
- **React.js** – Used for the development of the admin portal, providing a dynamic and responsive web interface for managing user data, bookings, and complaints.
- **Dart** – Primary programming language for Flutter, used to develop the logic and UI components of the mobile application.

ii. Backend Development :

- **Node.js** – Server-side JavaScript runtime used to build scalable backend services, handling API requests, business logic, and database interactions.
- **Express.js** – Web framework built on top of Node.js, used for creating RESTful APIs to facilitate communication between the frontend and the backend.
- **MongoDB** – NoSQL database used to store user data, booking details, complaint records, and bus form information.
- **Redis** – In-memory data store used to handle concurrency during bus bookings, ensuring efficient management of simultaneous booking requests.

iii. Payments and Third-Party Integrations:

- **PhonePe API** – Integrated to handle payments for bus bookings, allowing users to make secure payments directly from the app.
- **Firestore Cloud Messaging (FCM)** – Utilized for sending real-time push notifications to users' devices based on their subscription to specific topics related to bus forms.

iv. In-Memory Database (Redis):

- **Concurrency Control:** Redis manages concurrent bus bookings by ensuring atomic transactions, preventing double bookings and race conditions when multiple users attempt to book the same seat simultaneously.
- **Session Management & Data Expiry:** Redis stores session data for fast retrieval and automatically clears temporary booking information after a set duration, ensuring updated and efficient backend performance.

6.2 Results:

- i. Admins creates hostels and assign caretakers to designated hostels.

Create Hostel

Hostel Name

Number of Rooms

Create **Cancel**

Create Caretaker

Name:

Email:

Password:

Phone:

Hostel:

Create **Cancel**

Caretakers List

Add Caretaker

Name	Email	Phone Number	Hostel
Ganesh	caretaker@rguktn.ac.in	9876543210	I1 Hostel
Joshi Madugula	abc@gmail.com	9876543321	I2 Hostel

- ii. Admins and Caretakers can view issues raised by students

Issues Reported

Select Hostel:

Issue ID	Issue Type	Description	Student Name	Hostel Name
6800eda5a764925be2f2da63	CLEANING	room had to cleaned	Joshi	I1 Hostel
6800ae70cf0f98a43cae42cf	ELECTRICITY	Lights and fans are not working properly	Joshi	I1 Hostel

1

COMPLAINT DETAILS

Issue ID: 6800ae70cf0f98a43cae42cf

Type: ELECTRICITY

Description: Lights and fans are not working properly

Student Name: Joshi

Hostel: I1 Hostel

Priority: MEDIUM

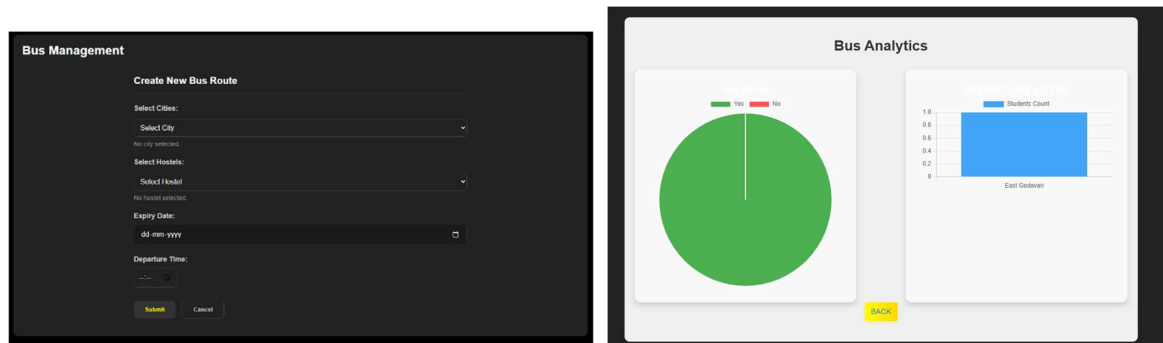
Status: PENDING

ATTACHED IMAGES:



Back to Issues

iii. Admins can create bus forms and can view analytics and create bus routes

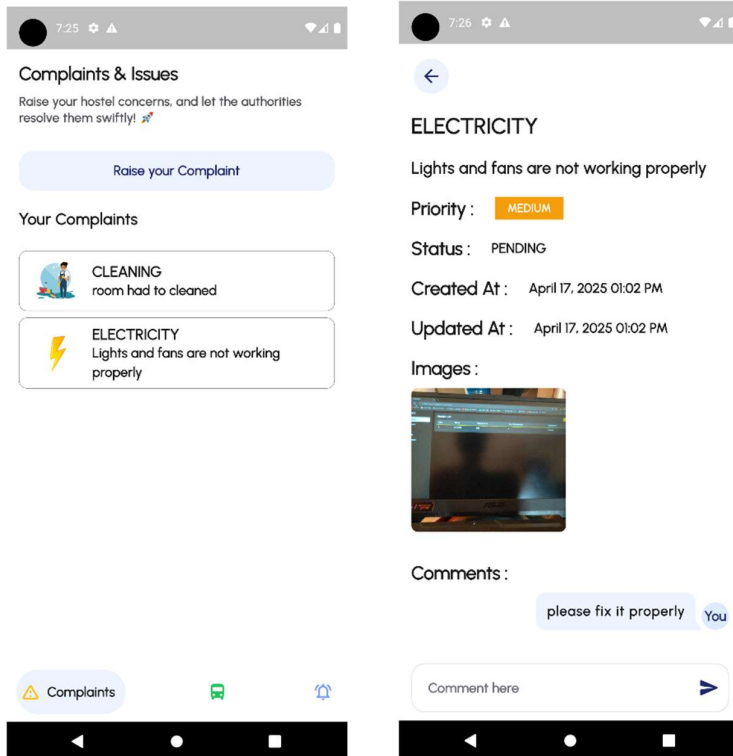


Passenger Name	Passenger Email	Passenger Phone	Booked By	Transaction ID	Amount	Payment Status
Joshi	n200395@rguktn.ac.in	8283837474	Joshi	TXN_1744881914571	1000	CONFIRMED
Joshi	n200395@rguktn.ac.in	9553329098	Joshi	TXN_1744875798715	1000	CONFIRMED

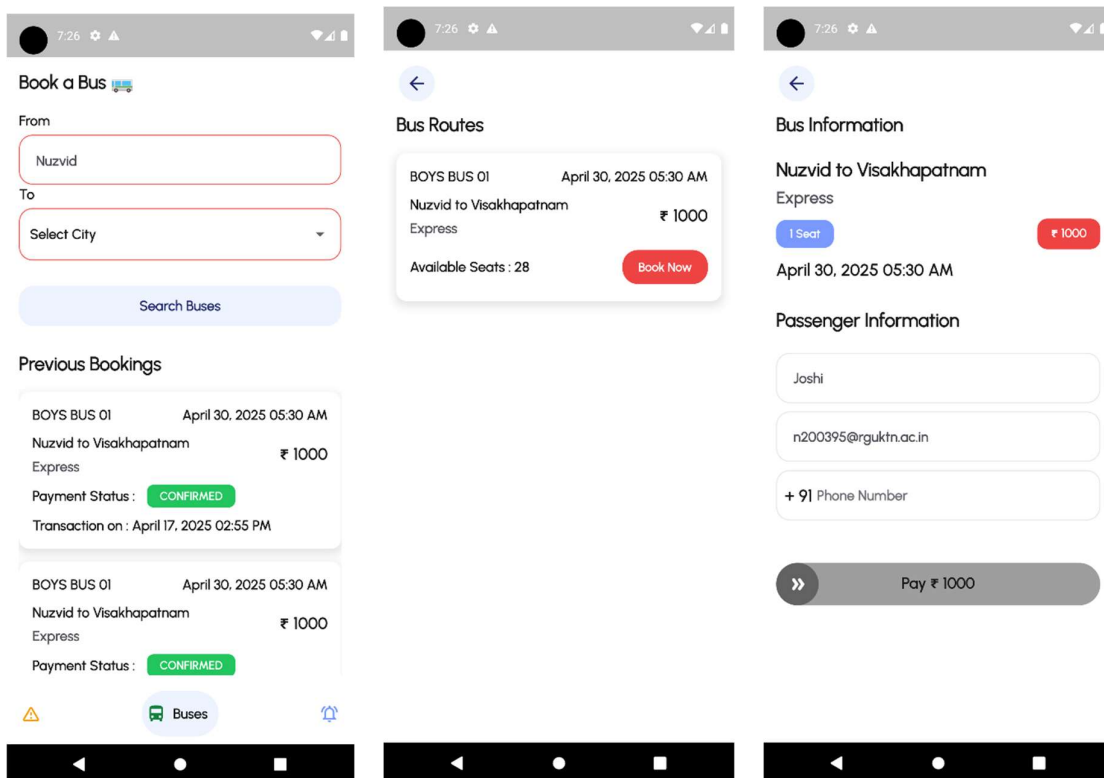
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Close

iv. Students can view their past complaints and raise a new complaint



v. Students can view past bookings, select city to view available buses and can book a bus



7. CONCLUSION

Our hostel monitoring system offers a comprehensive and efficient solution for managing hostel operations, ensuring seamless communication, and enhancing user experience. By integrating real-time monitoring, complaint management, bus bookings, and payment handling, the system streamlines administrative tasks while providing convenience for both students and hostel management. With a strong focus on security, concurrency control, and data integrity, the platform utilizes technologies like Redis for real-time performance, Firebase for notifications, and PhonePe API for secure payment processing. The use of a clean architecture, along with seamless integration of various modules, ensures that the system is both scalable and adaptable to future requirements. Ultimately, the hostel monitoring system redefines how hostels can be managed, providing a user-friendly and efficient digital solution that caters to the needs of both students and administrators.

8. FUTURE SCOPE

Future Scope of the Project :

i. Digital Outing Management:

- Currently, the process of managing student outings is manual, involving written permissions and paper slips. In the future, our system can fully digitize this process by incorporating QR codes for outing permissions. Students can submit requests digitally, and caretakers can approve or deny them through the portal. This will eliminate the need for physical documentation and streamline the entire process, making it more efficient and user-friendly.

ii. Smart Hostel Attendance System:

- The current attendance system for students in the hostel can be made smarter and more efficient by integrating machine learning-based face recognition into our app. This allows students to mark attendance simply by scanning their faces, enabling real-time, contactless, and accurate attendance tracking. It reduces manual effort, minimizes errors, and enhances security and convenience for both students and administrators.

iii. Expansion to All-Inclusive Hostel Management System:

- The platform can be expanded to cover all aspects of hostel management, turning it into a comprehensive software solution for managing hostel activities. This could include features like room allocation, maintenance requests, fee management, and mess tracking. Additionally, we can introduce a feedback system for students to rate their hostel experience, providing valuable insights for continuous improvement.

9. REFERENCES AND SOURCE CODE

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7. [Backend Code](#)
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